



CSE 350: Digital Electronics and Pulse techniques

Exp-04: Implementing Resistor Transistor Logic (RTL) and Diode Transistor Logic (DTL)
gates

Name:	Section:
ID:	Group:

Objectives

1. Constructing a Resistor Transistor Logic (RTL) gate.
2. Constructing a Diode Transistor Logic (DTL) gate.
3. Understanding the circuit operations.

Equipment and component list

Equipment

1. Digital Multimeter
2. DC power supply

Component

- NPN Transistor (C828) - x1 piece
- Diode 1N4003 - x2 pieces
- Resistors -
 - ❖ 2 K Ω - x2 pieces
 - ❖ 2.2 K Ω - x1 piece
 - ❖ 15 K Ω - x1 piece
 - ❖ 20 K Ω - x1 piece
 - ❖ 100 K Ω - x1 piece

Task-01: Inverter (NOT gate)

THEORY

In digital logic, an inverter or NOT gate is a logic gate which implements logical negation. It has a single input and a single output where the output is the exact opposite of the input. Meaning, if the input is Logical High, the output will be Logical Low and vice versa. The RTL implementation of an inverter circuit is shown in Figure 03.

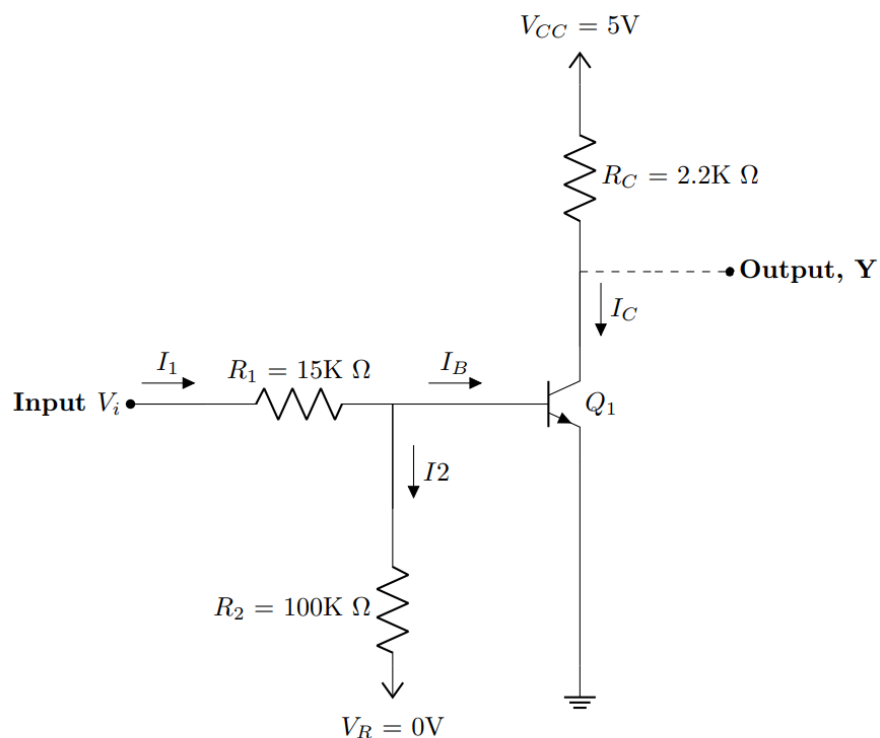


Fig 3: Inverter (NOT gate)

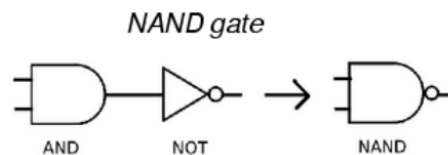
Here the input is applied to the base of a Transistor or, BJT (Q_1) through the resistor R_1 and the output is available at the collector terminal (point Y). We connect the ground terminal to the emitter node directly and to the base node through the resistor R_2 . Hence, when the input V_i is LOW (0V), the 'Base' terminal of the transistor cannot be at a voltage higher than zero. For Q_1 to be turned ON, the Base-Emitter voltage difference must be greater than 0.5V. Thus, the BJT cannot turn ON when the input is LOW and operates in cutoff mode. This means Q_1 acts like an open circuit and the current passing through the R_C resistor (I_C) is zero. As a result, there will be no voltage drop in the resistor R_C and the voltage of the output point (Y) will be the same as $V_{CC} = 5V$ (High).

On the other hand, if a HIGH input (5V) is applied at the input terminal (V_i), Q_1 will be driven into saturation mode. In this mode, the Collector-Emitter voltage difference (V_{CE}) is nearly 0.2V. As the emitter is connected to the ground terminal, the emitter voltage (V_E) is zero. Hence, the collector voltage will be close to 0.2V(LOW). Thus, the output of the circuit is always the opposite of the input.

Task-02: DTL NAND gate

THEORY

In this task, we will implement a Diode Transistor Logic (DTL) NAND gate. As can be seen in Fig. 4, a 2-input NAND gate outputs a logical HIGH if at least one of the inputs is LOW. Otherwise, the output of the NAND gate is logical LOW. It can be created by passing the output of an AND gate through an Inverter or NOT gate. One can build all other logic gates using such NAND gates. The DTL implementation of a NAND circuit is shown in Fig. 5.



A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

Figure 4: NAND gate Truth Table

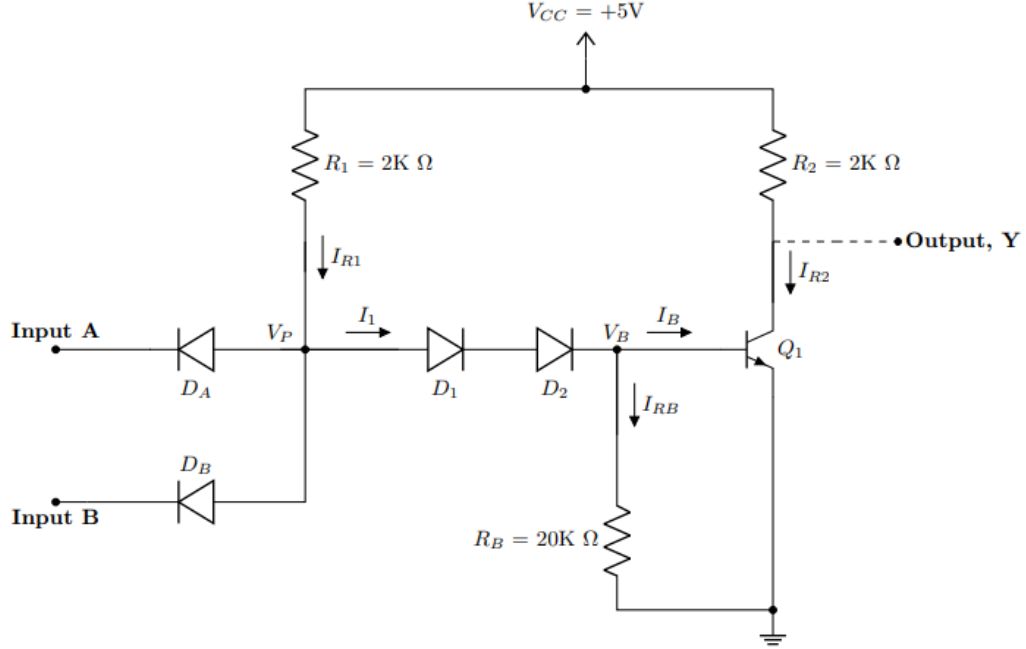


Figure 5: Diode Transistor Logic (DTL)

Diode–transistor logic (DTL) is a class of digital circuits that is the direct predecessor of transistor–transistor logic (TTL) and the successor of resistor–transistor logic (RTL). The output side of the basic DTL NAND circuit has a similar structure to the RTL inverter circuit, but it uses diode-logic AND on the input side. Thus, DTL circuits implement NAND gates by combining both DL and RTL logic, something that is not possible to do using solely RTL circuits or DL circuits. DTL also has better noise margin and fanout characteristics compared to RTL. (Noise margin indicates a circuit’s immunity to noise, and fanout is the number of identical gates that a circuit can drive without facing errors).

On the input side of Fig. 5, we can see two diodes in reverse bias connection with respect to input - this acts as an AND gate circuit according to diode logic. The output of this AND gate is fed to the inverter input V_B (base terminal of BJT Q_1) through the diodes D_1 and D_2 creating the NAND circuit. The output is obtained at the collector terminal of Q_1 .

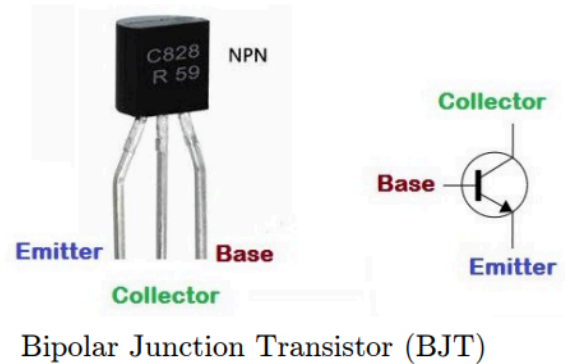
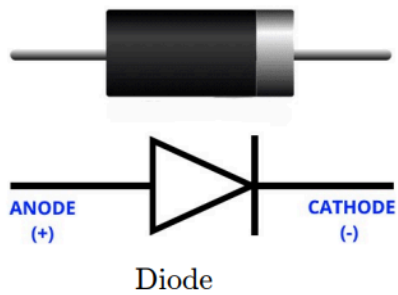
When both inputs are HIGH (5V), the cathode voltage of the diodes D_A and D_B become higher than their anode voltages. As a result, both input diodes operate in the reverse bias region (turned off/disconnected), and the node V_P has a high voltage level. This causes the transistor Q_1 to operate in the saturation mode and the NAND gate generates a LOW output. In this case, the voltage of point P (V_P) is close to 2.2V as the voltage of base terminal (V_B) in saturation is nearly 0.8V and there is a voltage drop of 0.7V in each of diodes D_1 and D_2 .

When either input is LOW (0V), the corresponding input diode operates in the forward bias (turned on), and there is a voltage drop of 0.7V across the diode. Hence, the voltage at the node V_P becomes only 0.7V higher than the LOW voltage at the input and drops much below the 2.2V required to turn on the BJT Q_1 . This causes the transistor Q_1 to work in the cutoff region and it acts like an open circuit. Hence, the current passing through the R_2 resistor (I_{R2}) is zero. As a result, there will be no voltage drop in the resistor R_2 and the voltage of the output point (Y) will be the same as $V_{CC} = 5V$ (High).

Procedure:

1. Connect the circuit as shown in Fig: 1, 2, 3 & 5.
2. Observe the output for all possible input combinations and thus verify the type of gate.
3. Fill up the following tables for OR gate, AND gate and Inverter.

The schematic and pin diagrams of diode and BJT are shown below:



Data Tables:

Table 1: RTL inverter Data

V_i (V)	V_{R1} (V)	V_{R2} (V)	V_{RC} (V)	V_Y (V)

Table 2: NAND Gate Data

V_A (V)	V_B (V)	V_P (V)	V_{R1} (V)	V_{RB} (mV)	V_{R2} (V)	V_Y (V)

Signature

Lab Tasks:

Lab Task-01

For the DTL circuit in Figure 5, Vary the **input A** from **0V to 5V** while keeping **input B** fixed at 5V. What is the **maximum** value of **input A** for which the **output** remains **HIGH**? [consider any voltage above 1V as **HIGH**]

Lab Task-02

For the DTL circuit in figure 5, remove the diodes D_1 and D_2 and replace them using a jumper wire. Then, measure the output voltage for the four different input cases. Verify and briefly explain if the circuit still works properly as a NAND gate.

V_A (V)	V_B (V)	V_Y (V)

Report

Please answer the following questions briefly in the given space.

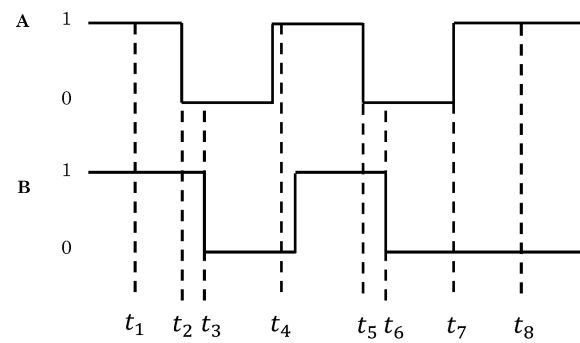
1. For the RTL circuit in figure 3, calculate the values I_1 , I_2 , I_C and I_B using your experimental data and then verify that the transistor will be operating in the saturation and cutoff region in two cases for the inverter circuit in figure 3 (Use experimental data for verification).

Ans.

2. What is the function of $V_R = 0V$ at the base of an inverter in figure 3?

Ans.

3. Assuming NAND gate, Draw the output.



4. Using experimental voltage data, calculate I_{R1} , I_B , I_{RB} , I_{R2} , find the **operating mode** of Q_1 in **figure 5 of the DTL** when both A and B are **HIGH**.

Ans.

5. Assume that the **output** of the circuit shown in Fig: 5 of the DTL is **LOW**. Draw the partial circuit consisting of only those components which remain active.

6. In a DTL NAND gate, when one input is held HIGH, its diode becomes reverse-biased and effectively removed from the circuit. Based on this, can the gate function as a DTL inverter (NOT gate)? Justify your answer using the behavior of the circuit and the NAND gate truth table.
7. Write a discussion and include the following: your overall experience, accuracy of the measured data, difficulties experienced, and your thoughts on those.